

INDEX

NAME Pooja AK SUBJECT _____
 STD. _____ DIV. _____ ROLL NO. _____ SCHOOL _____

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Completed
 Pooja

AIM:

To understand the basic syntax and features of the kotlin programming language.

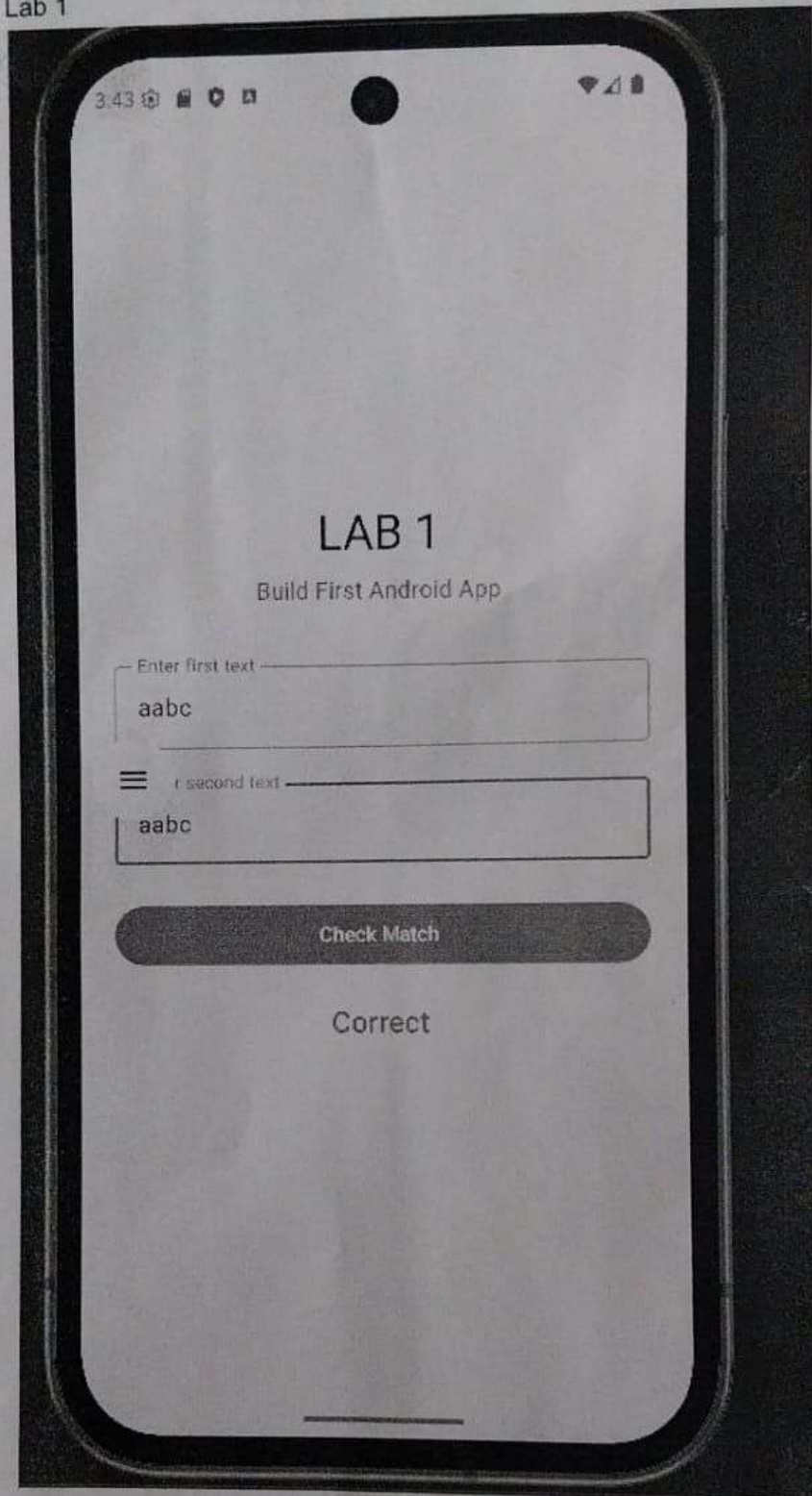
ALGORITHM:

- Start the android application
- Create a new android project in android studio using kotlin and Jetpack compose.
- Import the required packages and libraries
- Apply the application theme
- Run the application in android Emulator or on physical device.
- Verify output
- Stop the program.

RESULT:

The basic Kotlin program was implemented successfully & fundamental syntax & operation understood.

Lab 1



Functions

AIM:

To study and implement functions in Kotlin for performing specific tasks.

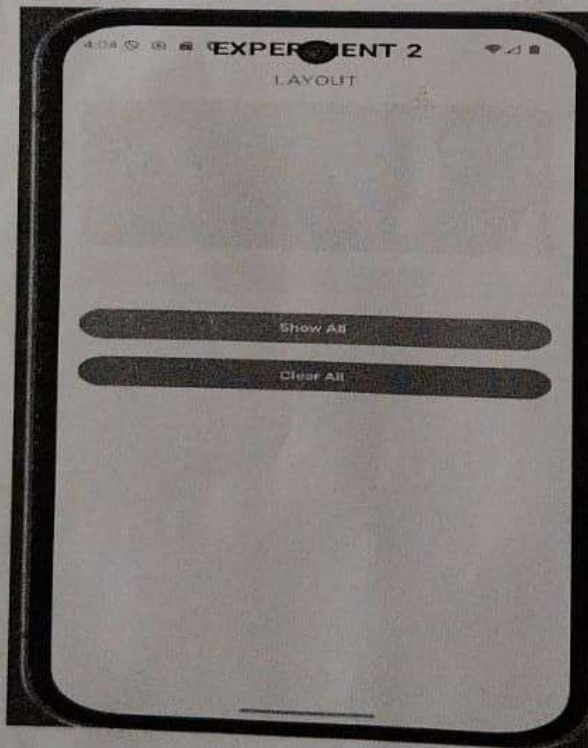
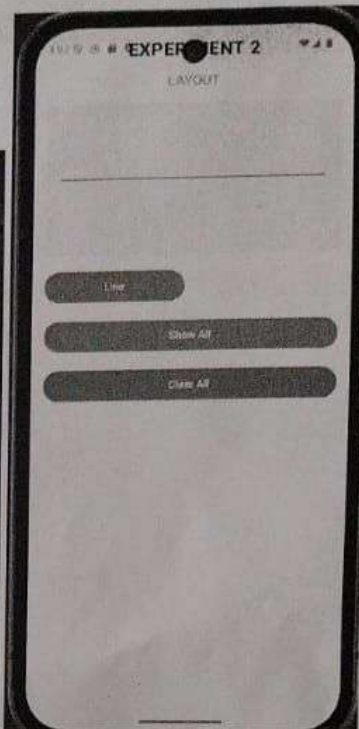
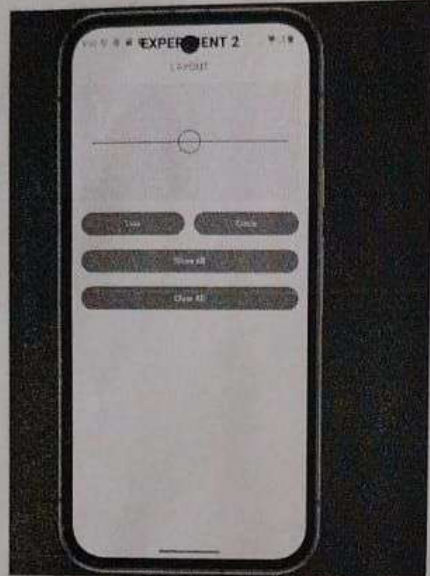
ALGORITHM:

- Start the Algorithm
- Create a new Android project in Android Studio using Kotlin & Jetpack compose
- Import the required packages and libraries.
- Apply the application theme.
- Implement the required UI functionality using State Variables and eventhandlers.
- Run the application in Android emulator
- Verify output

RESULT:

Function were created and execute Successfully demonstrating modular programming in Kotlin.

Lab2



Handwritten notes at the bottom of the page, likely describing the app's functionality or the lab exercise.

DATE:

AIM:

To understand object oriented programming concepts using classes and objects in Kotlin.

ALGORITHM:

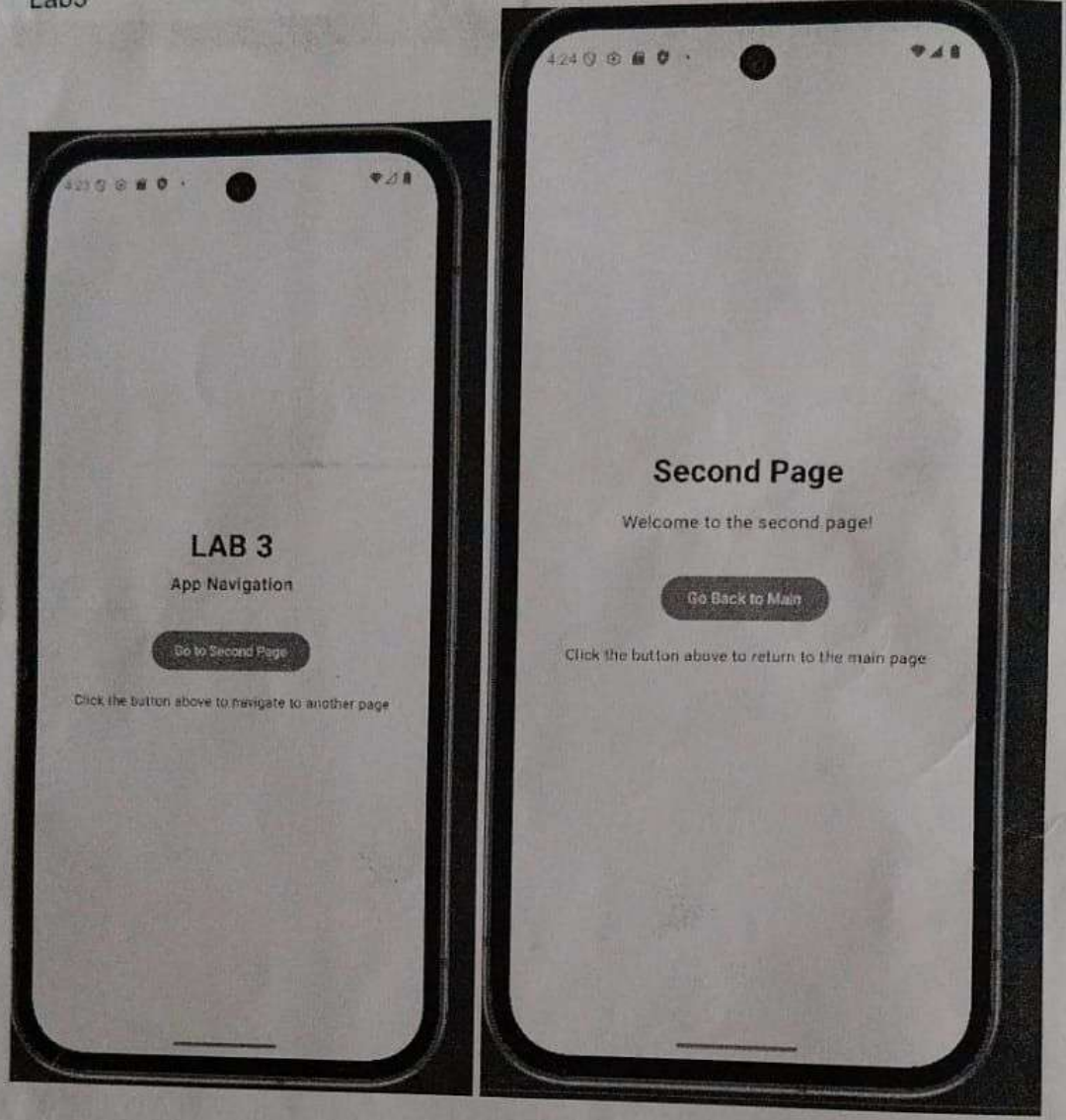
- Start the Android application
- Create a new Android project in Android Studio using Kotlin & Jetpack compose
- Import required packages and libraries
- Apply application theme
- Run the application in an Android emulator or physical device
- Verify the output
- Stop the program.

RESULT:

Classes and objects are created successfully and the concept of object-oriented programming was understood.

By

Lab3



DATE:

AIM:

To create and run a simple Android application using Android Studio and Kotlin

ALGORITHM:

- Start the Android application
- Create a new project in Android Studio using Kotlin and Jet compose
- Import required package and libraries
- Apply the application theme
- Run the Application in an Android Emulator or physical device
- Verify the output
- Stop the program

RESULT:

A basic Android application was created and executed successfully in emulator / device.

AIM:

To design and implement different layouts in an Android Application using Jetpack compose.

ALGORITHM:

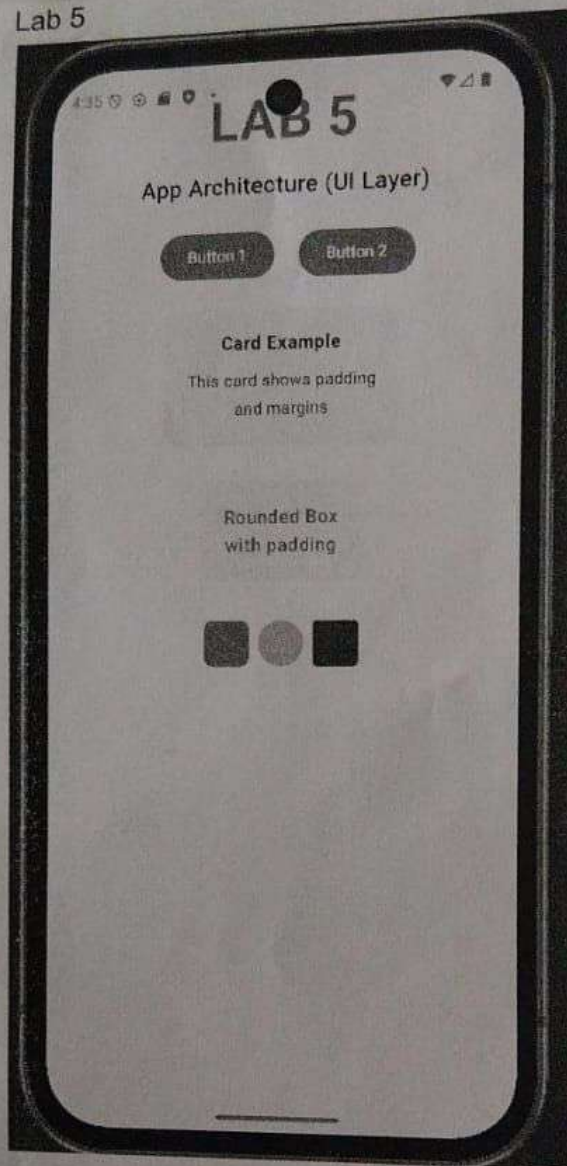
- Start the Android Application
- Create a new Android project in Android studio using Kotlin and Jetpack Compose.
- Import the required packages and libraries
- Apply the application theme
- Implement the required UI functionality using state variables & event handlers.
- Run the application in an android emulator.
- Verify the output.

RESULT:

Layout were successfully designed and UI components were arranged properly in the application.

800

Lab 5



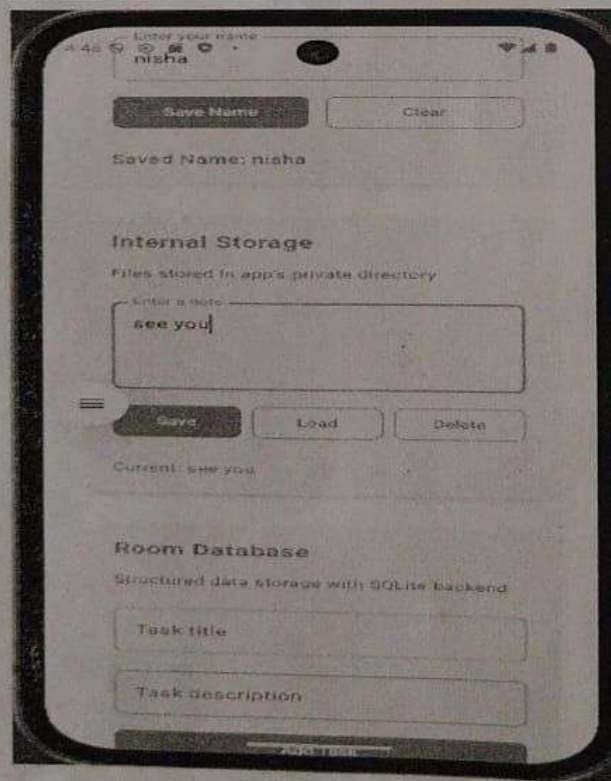
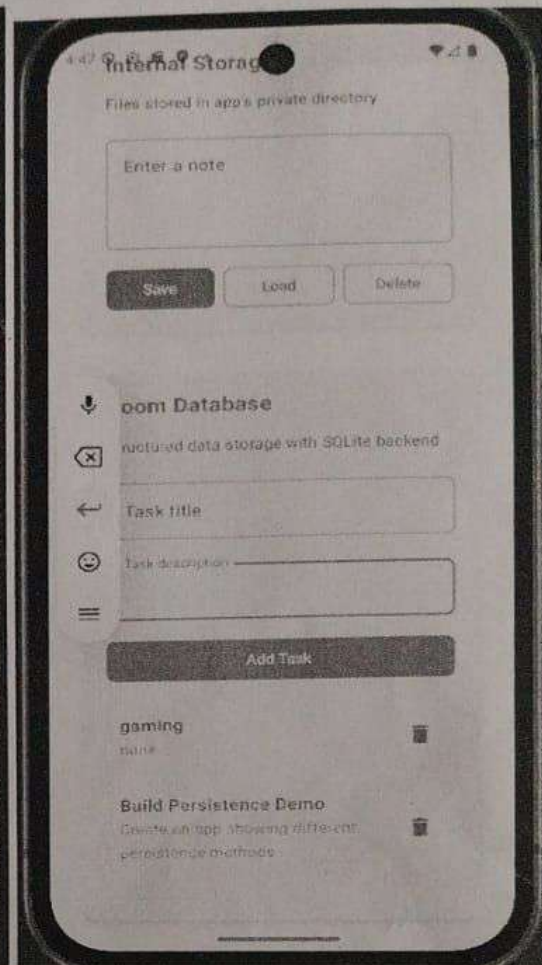
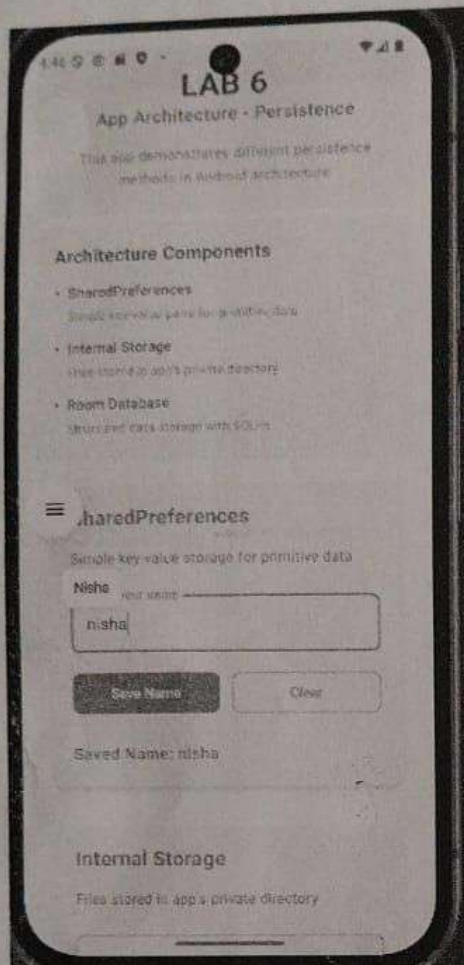
AIM: To implement app architecture between different screens in an android application

ALGORITHM:

- Start the Android Application
- open an android studio & create a new project using kotlin & Jetpack Compose.
- Add Navigation compose dependency in project
- Create different screens.
- Initialize a NavController to manage navigation between screens.
- Define navhost & specify routes for each screen
- Run the application on emulator
- verify the output.

RESULT:

App architecture persistence multiple screens was implemented successfully



DATE:

AIM:

To Study the advanced recyclerview use cases in Android.

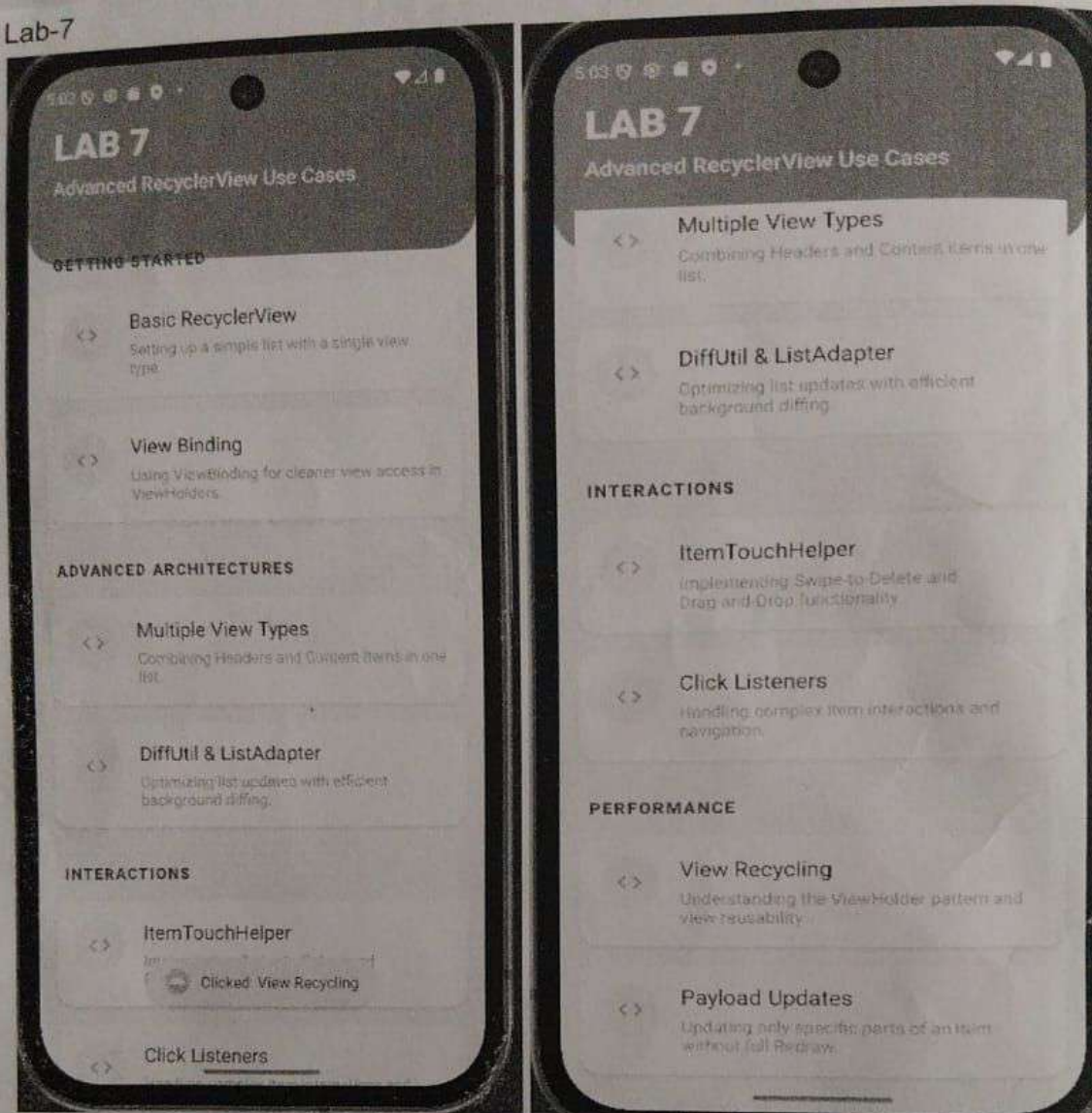
ALGORITHM:

- Start the android application & open android studio to create a new kotlin project
- Add the RecyclerView dependency & design a layout containing a recycler view.
- Create a data model & an adaptor with View Holder to display the list items
- add items interactions such as click listener & swipe actions using Item user Interfaces.
- Run the application & verify output.

RESULT:

The advanced recyclerview use cases was observed and understood successfully.

Lab-7



AIM:

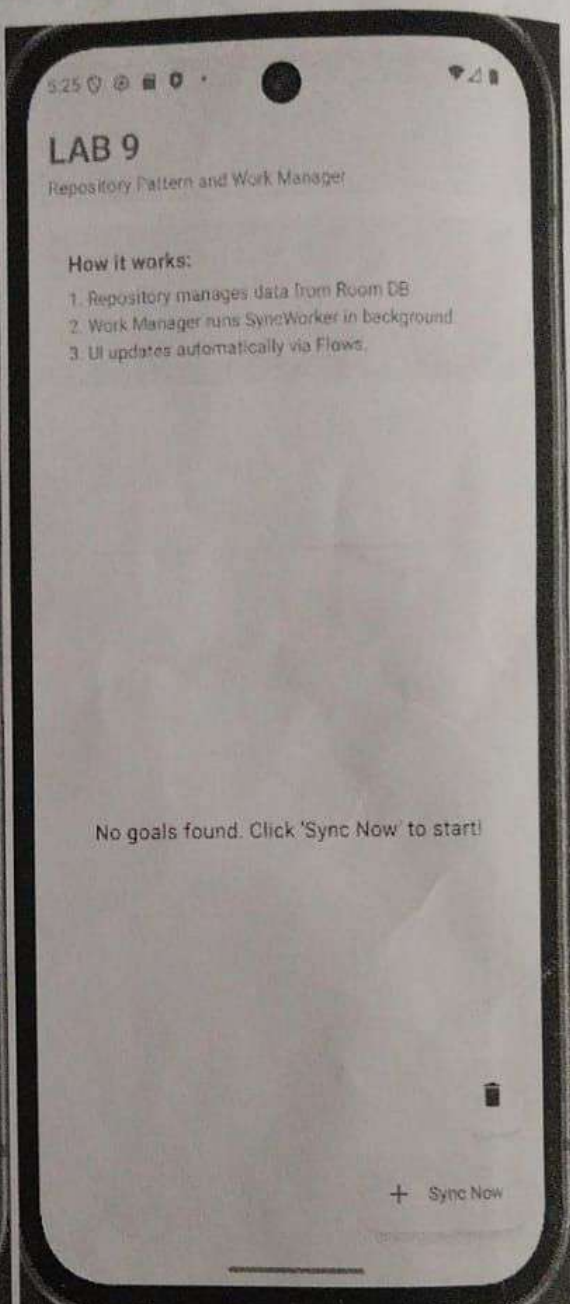
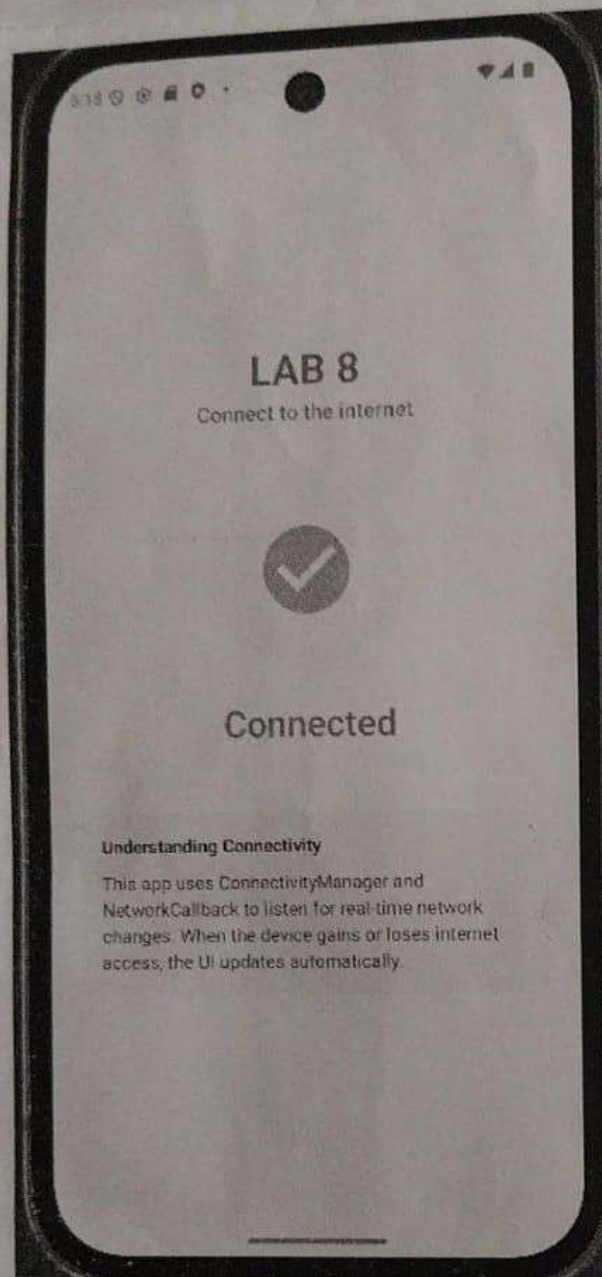
To connect the Android application to the Internet & retrieve data from online sources.

ALGORITHM:

- To connect the Android Application to the Internet and retrieve.
- Start with Android App & Create a new project in Android Studio using Kotlin.
- Use Connectivity Manager to monitor the network connection status.
- Update the UI to display the connection status.
- Run the application & verify the status change.

RESULT:

The application successfully connected to Internet and receive required data.



AIM:

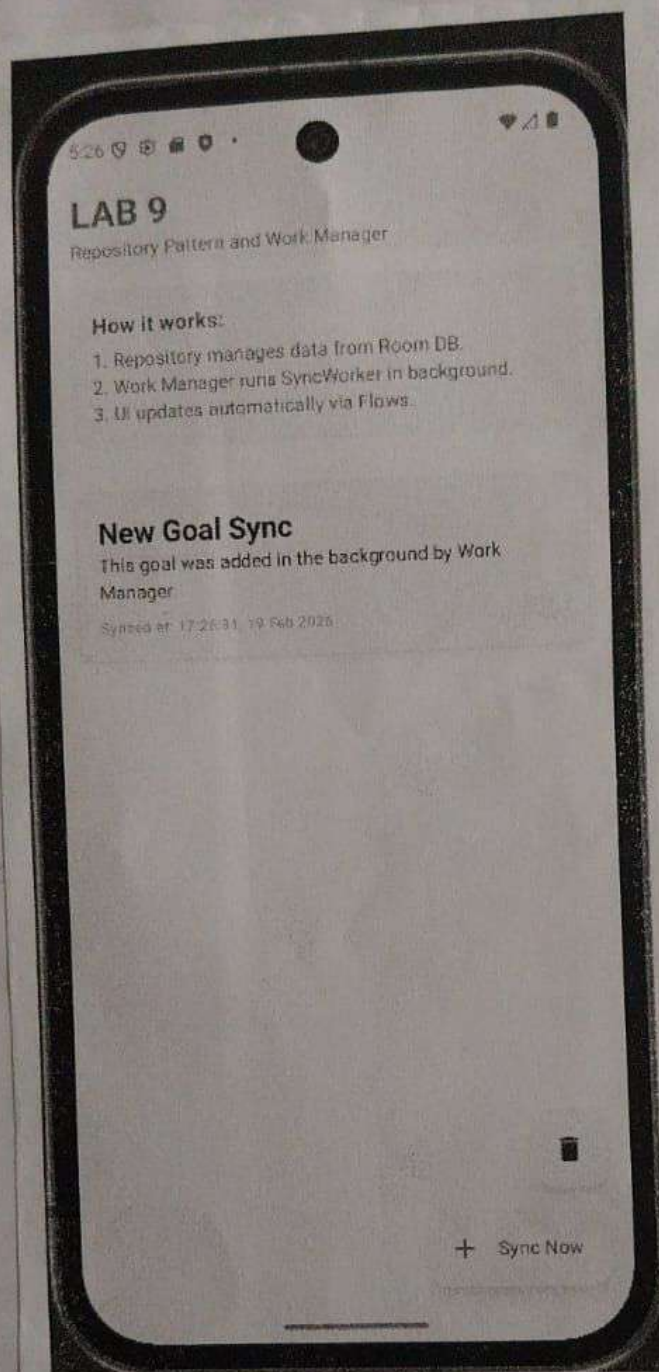
To implement the repository pattern & workmanager for background tasks in Android.

ALGORITHM:

- Start the Android Application and create a new project in Android Studio using Kotlin.
- Create a Room database and implement an repository pattern to manage data operations.
- Connect the repository with worker to share and update data in database.
- Run the application & verify.

RESULT:

Repository patterns and workmanager were implemented successfully for data management & background processing.



Ex No: 10

DATE:

APP UI DESIGN

19

AIM:

To design an attractive and user-friendly Interface for an Android Application.

ALGORITHM:

- Start the Android application & create a new project in Android studio using Kotlin.
- Design the main UI layout using components such as buttons text.
- Implement UI design principle
- Add interactive components
- Apply proper alignment
- Run the application.

RESULT:

The application UI was designed successfully with proper layout & styling.

